

Inositol hexaphosphate (IP6) inhibits key events of cancer metastasis: II. Effects on integrins and focal adhesions.

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BACKGROUND: We have shown that inositol hexaphosphate (IP6), a natural compound and a potent anti-cancer agent, inhibited cancer cell adhesion to the extracellular matrix (ECM) proteins, thereby leading to inhibition of cell migration and invasion. Cell adhesion to ECM is mediated by specific cell surface integrins, which transduce intracellular signals through their interaction and activation of other proteins that are recruited to the focal adhesion. We hypothesize that IP6 decreases cell adhesion by suppressing the integrin receptors and their subsequent signaling pathway. **MATERIALS AND METHODS:** We analyzed integrin expressions of the highly invasive estrogen receptor-negative human breast cancer MDA-MB 231 cells exposed to IP6 by flow cytometry. The expression of focal adhesion proteins was investigated by immunocytochemistry and Western blotting. **RESULTS:** IP6 treatment caused a significant ($P < 0.005$) decrease in the expression of integrin heterodimers alpha 2 beta 1 (collagen receptor), alpha 5 beta 1 (fibronectin receptor) and alpha v beta 3 (vitronectin receptor); flow cytometry showed that it was the alpha 5 subunit that was down-regulated (< 0.001). However, the expression of the alpha 2, alpha v, beta 1 and beta 3 subunits were not affected by IP6 treatment. When the expression of integrins on the cell surface was assessed, there was a dramatic 82% decrease in the expression of alpha 5 beta 1 on IP6-treated cells ($P < 0.0001$), indicating a decrease in cell surface expression of the heterodimers. No effect was seen when inositol hexasulfate (IS6), an analogue of IP6, was used as a control. Immunocytochemistry showed a lack of clustering of paxillin; tyrosine-phosphorylated proteins in IP6-treated cells were discontinuous and scattered around the cell periphery, whereas the patterns were more dense and localized in control cells. Consistent with these observations, focal adhesion kinase (FAK) autophosphorylation at tyrosine-397 residue was suppressed, albeit modestly, by IP6 treatment, suggesting a down-regulation in the integrin-mediated signaling pathway. **CONCLUSION:** The results of this study indicate that IP6-induced inhibition of cancer cell adhesion, migration and invasion may be mediated through the modulation of integrin dimerization, cell surface expression and integrin-associated signaling pathway.